

SOFTWARE TEST PLAN & TESTING DOCUMENTATION

RAIMES - Responsible AI Mining Evaluation System

1. TEST PLAN OVERVIEW

1.1 Project Description

RAIMES is an AI-based evaluation system for assessing responsible mining practices. It allows mining companies to conduct self-assessments and receive AI-generated evaluations of their sustainability and compliance practices.

make it in english

1.2 Test Objectives

- Verify system functionality according to requirements
- Ensure data security and user authentication
- Validate system performance under normal and high load conditions
- Test cross-platform and browser compatibility
- Ensure proper integration between components

1.3 Test Scope

In Scope:

- Authentication system (login, register, JWT)
- Question management (CRUD operations)
- Database operations and connections
- API endpoints and responses
- Frontend user interface
- CORS configuration
- Real-time search and filtering

Out of Scope:

- AI validation service (not yet implemented)
- Evidence upload functionality (future feature)
- Email notifications (future feature)
- Mobile application testing

1.4 Test Approach

- **Unit Testing:** Testing individual components/functions
- **Integration Testing:** Testing component interactions
- **System Testing:** End-to-end testing
- **User Acceptance Testing:** Business requirement validation

- **Performance Testing:** Load dan stress testing
 - **Security Testing:** Authentication dan authorization
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2. TEST STRATEGY

2.1 Testing Levels

2.1.1 Unit Testing

- **Framework:** Jest (Frontend), Mocha/Chai (Backend)
- **Coverage Target:** 80% minimum
- **Focus:** Individual functions, components, API endpoints

2.1.2 Integration Testing

- **API Integration:** Postman/Newman
- **Database Integration:** SQL test scripts
- **Frontend-Backend Integration:** Cypress/Selenium

2.1.3 System Testing

- **End-to-End:** Complete user workflows
- **Cross-browser:** Chrome, Firefox, Safari, Edge
- **Responsive Design:** Desktop, tablet, mobile viewports

2.1.4 Performance Testing

- **Tools:** JMeter, Lighthouse
- **Load Testing:** 100 concurrent users
- **Response Time:** < 2 seconds average

2.1.5 Security Testing

- **Authentication Testing:** JWT validation, password security
- **Authorization Testing:** Role-based access
- **OWASP Top 10:** SQL injection, XSS prevention

2.2 Testing Types

2.2.1 Functional Testing

- Feature functionality validation
- User interface testing
- API functionality testing
- Database operations testing

2.2.2 Non-Functional Testing

- Performance testing
 - Security testing
 - Usability testing
 - Compatibility testing
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3. TEST SCOPE

3.1 Features to be Tested

3.1.1 Authentication Module

- User registration
- User login
- JWT token generation dan validation
- Password hashing
- Protected route access
- Session management

3.1.2 Question Management Module

- Create new questions (essay/multiple choice)
- Read/View questions list
- Update existing questions
- Delete questions
- Search functionality
- Filter by category dan type
- Weight assignment validation

3.1.3 Database Module

- Connection pooling
- CRUD operations
- Data validation
- Error handling
- Transaction management

3.1.4 API Module

- RESTful endpoints
- Request/response validation
- Error handling
- CORS configuration
- Rate limiting (future)

3.2 Features NOT to be Tested

- AI validation service (not yet implemented)
- Evidence upload (future feature)

- Report generation (future feature)
 - Email notifications (future feature)
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4. TEST ENVIRONMENT

4.1 Hardware Requirements

- **Development Machine:** Windows 11, 16GB RAM, Intel i7
- **Test Server:** Ubuntu 20.04 LTS, 8GB RAM
- **Database Server:** Supabase Cloud PostgreSQL

4.2 Software Requirements

- **Operating Systems:** Windows 11, macOS, Ubuntu 20.04
- **Browsers:** Chrome 119+, Firefox 119+, Safari 17+, Edge 119+
- **Node.js:** v18.x or higher
- **PostgreSQL:** v15+ (Supabase)

4.3 Network Configuration

- **Frontend:** http://localhost:5174 (not deployed yet)
- **Backend API:** http://localhost:3000 (not deployed yet)
- **Database:** Supabase cloud connection
- **CORS:** Multi-origin support enabled

4.4 Test Data

- **Users:** 10 test accounts with different roles
- **Questions:** 50+ sample questions (essay/multiple choice)
- **Categories:** Environmental, Safety, Social, Governance
- **Test Database:** Separate from production